

# Second Look

## A two-minute test of how well a volunteer sees a creek, and a model that may only ask

Alex Velazquez and Rachel Selbrede, September 2026

Live: <https://second-look-79t.pages.dev> · Code: <https://github.com/alejandro-publius/second-look>

### Abstract

Citizen observations of creeks are uneven. People notice smell, foam and colour and walk past concrete banks, dug-out channels, plants that do not belong and pipes. Second Look measures that, per person and per feature, with a two-minute test on 16 real photos, and stores the score with every observation the person later makes, in OneAquaHealth's own FHIR profiles. Vision models take the same test. A model may only ask a volunteer to look again, on a feature it passed, after the person has answered; code, never the model, writes the record. Four Claude models took the test three times each. Every model passed built banks, and no model passed plants that do not belong, where almost every answer was "can't tell": the plant photos show no water, and the instruction says to answer can't tell when no stream is in view. On 46 frames of open creek footage the gate dropped 29 of 64 candidate flags, each on a feature that model had not passed. The HL7 validator found 0 errors in 17 records checked against OneAquaHealth's guide. No person has taken the test yet, and this report makes no claim about people; a paid research panel may add sessions before the lock, and they would be reported once, after it.

### 1. The problem

People judge a creek the way they judge a park. Tidy and green reads as healthy. OneAquaHealth's project lead said it in the first workshop: volunteers catch smell, foam and colour, and walk past concrete banks, a channel that was dug out, and pretty plants that do not belong. So the best-looking creek can get the best rating and deserve the worst.

Professional surveyors fixed this long ago. In the UK's River Habitat Survey, "only surveys from accredited surveyors will be entered on the RHS database", and accreditation means attending a course and passing a test (RHS manual 2003, pages 3 and 20; see [docs/notes/sources.md](#)). Some volunteer programs certify people for a method, such as water chemistry. None we know of measures how well each volunteer sees each feature, or keeps that score with every observation. So a volunteer's observation arrives with no mark of how well its observer sees.

Measure each volunteer, per feature, and store the measure with the data. The analyst sees "4 of 4 on built banks, tested Sep 23" beside an answer, never a blended grade or a probability. Follow-up questions are chosen by code from the answers, the person's scores and the weather, two at most: "It has not rained here for N days. Is anything coming out of that pipe?" The AI takes the same test as the people and earns the right to ask one question, feature by feature.

*Every risk, what stops it, and the test that proves it*

| What can go wrong                               | What stops it, by construction                                                                                                                                                        | Proof                                                                                                                                                  |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| A volunteer walks past a built bank             | The lesson teaches the four features people miss, and the test measures each one; the score is stored with every answer                                                               | <code>content/lessons/, core/scoring.py, core/tests/test_scoring.py</code>                                                                             |
| Someone taps at random                          | Four items per feature, two present and two absent, so chance scores about 2 of 4; a low scorer who answers No is asked for a photo                                                   | <code>content/test_items.yaml, core/followups.py</code> rule <code>low_score</code>                                                                    |
| A pretty creek gets the best rating             | A good rating beside a reported built bank, sewage or plant that does not belong triggers one question asking the person to keep or change it, in the creek check and in a video walk | <code>content/followups.yaml</code> rule <code>rating_check</code> , <code>core/tests/test_followups.py</code> , <code>core/tests/test_walks.py</code> |
| A pipe report means nothing without the weather | The dry pipe question fires only after dry days from Open-Meteo, and is skipped when the weather is unknown                                                                           | <code>core/rainfall.py, core/tests/test_rainfall.py</code>                                                                                             |
| The model invents a feature                     | Model output becomes a Flag through the gate or is dropped; a fuzz test throws arbitrary output at it                                                                                 | <code>core/gate.py, core/tests/test_gate.py</code>                                                                                                     |

| What can go wrong                                  | What stops it, by construction                                                                                                                                                                                                                                                                                                                                                                                                 | Proof                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The model was never good at that feature           | A model may flag only a feature it passed on the same test as the people; the pass table is a committed file, and one from the fake client licenses nothing                                                                                                                                                                                                                                                                    | <code>results/model_pass_table.json</code> , <code>core/tests/test_checker.py</code>                                                                                                                                                                                                                                                                                                                   |
| Duplicate and test pins fill the map               | A precise pin within 30 metres of an existing spot is offered as that spot; names that read like a test are kept out of the counts and listed apart for a person to check; a coarse pin is never compared                                                                                                                                                                                                                      | <code>core/act.py</code> , <code>core/tests/test_act.py</code>                                                                                                                                                                                                                                                                                                                                         |
| A record a city's systems cannot read              | Sample records from both emitters, the Python API and the TypeScript Worker (17 files, walk records among them), are validated against their guide in CI, and golden vectors hold the live emitter to the same output                                                                                                                                                                                                          | <code>scripts/fhir_validate.py</code> , <code>worker/test/golden.test.ts</code>                                                                                                                                                                                                                                                                                                                        |
| Someone edits history                              | A hash-chained audit log, checked by a script                                                                                                                                                                                                                                                                                                                                                                                  | <code>audit/</code> , <code>scripts/verify_audit.py</code>                                                                                                                                                                                                                                                                                                                                             |
| Someone rewrites the audit log, last line included | Once a day its last hash is stamped with OpenTimestamps, if it has changed since the last stamp. A rewrite with fresh hashes still holds together, so the stamped line is compared with the log's own line; <code>/verify</code> does that too, and checks the whole chain again in your browser                                                                                                                               | <code>uv run python scripts/verify_audit.py</code> , then <code>uv run python scripts/ots_status.py</code> , which fails when the log no longer has a stamped line; <code>.venv/bin/ots verify proofs/audit-head-2026-09-24.ots</code> checks the stamp itself                                                                                                                                         |
| The plan was written after the data came in        | The <code>prereg-v1</code> tag object and <code>docs/analysis_plan.md</code> are stamped with OpenTimestamps, a public timestamp service that anchors hashes in Bitcoin; it is not our own chain. <code>/verify</code> shows each proof and its Bitcoin block once confirmed                                                                                                                                                   | <code>.venv/bin/ots verify proofs/prereg-v1.tag.ots</code> , <code>.venv/bin/ots verify -f docs/analysis_plan.md proofs/analysis_plan.md.ots</code> ( <code>uv sync</code> puts <code>ots</code> in <code>.venv/bin</code> ; it needs a Bitcoin node to finish), or <code>uv run python scripts/ots_status.py</code> , which checks the block against a public explorer; <code>proofs/README.md</code> |
| We fool ourselves with the statistics              | The analysis plan is tagged before any data; every README number is checked against <code>results/</code> in CI; a synthetic result is cited only where the sentence says it is a simulation                                                                                                                                                                                                                                   | <code>docs/analysis_plan.md</code> at <code>prereg-v1</code> , <code>scripts/verify_claims.py</code>                                                                                                                                                                                                                                                                                                   |
| Judge mode leaks the answer key                    | Before the lock, this: judge mode is shut until Sep 28 by a lock constant, and its answer route refuses before then on the Worker and in the Python API. After the lock, nothing: judge mode says right or wrong on the same 16 photos the test uses, so anyone can rebuild the key and carry a perfect score into their creek checks, and a retake in a fresh browser does the same at any time. Known weaknesses says so too | <code>core/lock.py</code> , <code>apps/api/tests/test_study.py::test_demo_answer_is_shut_before_the_lock</code> , <code>worker/test/e2e.mjs</code> ; what is not stopped: <code>docs/THREAT_MODEL.md</code>                                                                                                                                                                                            |
| A refresh loses a session                          | The session resumes from the server; the creek check queues offline and sends later                                                                                                                                                                                                                                                                                                                                            | <code>apps/web/lib/offline.ts</code> , <code>apps/web/tests/</code>                                                                                                                                                                                                                                                                                                                                    |

## 2. Method

How a model's answer becomes a flag, or nothing, is drawn in `docs/diagrams/ai-gate.svg`. It is too wide to print here with text you can read, and 2.2 gives the same steps in words.

### 2.1 The test

Sixteen photos, four per feature, two with the feature and two without, in a new random order for each person. The answers are Yes, No and Can't tell, and Can't tell counts as wrong. The score is kept per feature, out of 4, and stored in the record of the test sitting; a dated qualification for the test and a Provenance link carry it to every Observation the person makes. The question wording and the analysis were written into `docs/analysis_plan.md` and tagged `prereg-v1`, and the key was frozen as a hash in `results/key_hash.json`, on Sep 21, the day the test opened, before the first paid model run on Sep 23. The dates here are Pacific time.

### 2.2 The gate

A vision model can help a volunteer look again. It can never decide what is stored. Every model answer takes this path:

1. **The person answers first.** The creek check asks the official app's questions in its order (`content/form.yaml`), and the answers are the person's own. The questions and answers are the app's own words, quoted from its public translation bundle and checked again by `make app-strings-check`.

2. **The model is asked only where it passed.** `core/checker.py` reads the committed pass table, `results/model_pass_table.json`, and does not even ask about a feature that model did not pass on the same 16-photo test the volunteers take. A table that does not say `"real": true` licenses nothing.
3. **Its answer is forced.** One photo, one feature, the frozen question wording. Anything that is not yes, no or can't tell with a short note becomes can't tell and counts as malformed (`force_answer` in `core/checker.py`).
4. **The gate turns it into a flag or drops it.** `parse_flags` in `core/gate.py` keeps a `Flag` only for a known feature the model passed, with a sane confidence and a short plain note, and drops everything else with a reason in plain words. It never raises. On the footage run it dropped 29 of 64 candidate flags, each for a feature that model had not passed.
5. **A flag can only make one question eligible.** `core/followups.py` picks the follow-up questions from the answers, the rain (`core/rainfall.py`), the person's scores and the flags, by the rules in `content/followups.yaml`: two questions at most, the model's at most one, and no model call inside it.
6. **The person taps "I looked again" or "Skip", and no stored answer changes.** The question comes only for a feature the check asks about, and the model's note is shown only as "the checker noticed" (`core/followups.py`, `docs/MODEL_CARD.md`).
7. **The record is built from human inputs only.** `build_record` in `core/gate.py` has no parameter that could carry a flag, a model id or model text. `core/fhir_emit.py` writes the record under OneAquaHealth's profiles with the person's score attached, and `scripts/fhir_validate.py` checks it in CI.

Where it runs today: nowhere a person sees yet. `scripts/build_walks.py` sends the footage run's answers through the same gate when the video walks are built, and no flag on those clips passed it, so 0 of the 3 walks shows a checker question. The live creek check runs with the checker off (`CHECKER_ENABLED`), so both servers pass it no flags and no model is called. `examples/footage-flag/` shows it act on the footage run.

## 2.3 What the AI cannot do

It cannot write the record. It cannot speak on a feature it did not pass, or on a made-up pass table. It cannot ask more than one question, or ask before the person answers. The gate says how, and Three properties that follow names the test for each. Two more limits:

| It cannot                                                | Enforced by                                                                                                                                                    | Test                                                                                                                                                                           |
|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Put more than a short labelled note in front of a person | The gate drops a note with markup, line breaks or text direction controls, and the page shows the note only after "the checker noticed", cut to 160 characters | <code>core/tests/test_gate.py::test_note_with_markup_or_a_direction_control_is_dropped</code> , <code>core/tests/test_checker.py::test_long_note_is_cut_to_160_and_flat</code> |
| State a risk for a named site                            | Every health or ecology sentence comes from <code>content/approved_sentences.yaml</code> with a source                                                         | <code>core/tests/test_healthcard.py</code> , <code>core/tests/test_act.py</code>                                                                                               |

## 2.4 Three properties that follow

| Property                                                                                       | Why it holds                                                                                 | The test that fails if it breaks                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The model cannot write the record                                                              | The only way to make a record takes no model output                                          | <code>core/tests/test_gate.py::test_build_record_signature_carries_human_inputs_only</code> , and the fuzz test <code>core/tests/test_gate.py::test_fuzz_model_output_never_reaches_answers_or_labels</code>                                                               |
| A model speaks only where it passed, and a made-up pass table licenses nothing                 | The gate and the checker both read the committed table and require <code>"real": true</code> | <code>core/tests/test_harden_gate_properties.py::test_every_kept_flag_is_licensed_for_that_exact_model</code> , <code>core/tests/test_checker.py::test_synthetic_pass_table_never_licenses_a_flag_by_default</code>                                                        |
| Code chooses the questions: two at most, the model's at most one, no network and no model call | Follow-up selection is a pure function                                                       | <code>core/tests/test_harden_followups_properties.py::test_the_content_table_never_asks_more_than_two_questions_for_any_input</code> , <code>core/tests/test_harden_followups_properties.py::test_the_selector_runs_with_http_and_the_model_client_patched_to_raise</code> |

The hard parts of building this, and how each is proved, are in `WRITEUP.md`. The decisions are in `docs/adr/`.

### 3. Evidence

All numbers below come from files in `results/`, written by the scripts in `evals/` and checked against the text by `scripts/verify_claims.py` in CI. Failed runs stay in `results/` too.

#### 3.1 The models on the test and on footage

|                                                                                                                                                           | The 16-photo test people take                                                                                                                                                                                                    | Frames from open creek footage                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pool                                                                                                                                                      | 16 photos, 4 per feature, the frozen question wording                                                                                                                                                                            | 46 frames from 5 openly licensed videos in 3 countries, screened for people and text                                                                                            |
| Right answers, three runs of the 16 photos                                                                                                                | Claude Haiku 4.5 31 of 48, Claude Sonnet 5 33 of 48, Claude Opus 5.5 33 of 48, Claude Fable 5.1 34 of 48; per feature, with intervals, in <code>results/benchmark_20260924T054939Z.json</code>                                   | not measured: none of the frames has a label, so this column reports agreement                                                                                                  |
| A checker that always answers No, the floor to beat                                                                                                       | 24 of 48                                                                                                                                                                                                                         | not asked                                                                                                                                                                       |
| Right answers without the plant photos                                                                                                                    | Claude Haiku 4.5 31 of 36, Claude Sonnet 5 33 of 36, Claude Opus 5.5 33 of 36, Claude Fable 5.1 34 of 36                                                                                                                         | not asked                                                                                                                                                                       |
| Which features each model passed                                                                                                                          | the table below                                                                                                                                                                                                                  | not asked: footage decides nothing, a pass is earned on the test                                                                                                                |
| Agreement between models                                                                                                                                  | not asked                                                                                                                                                                                                                        | Haiku 4.5 and Fable 5.1, the pair that agreed least, gave the same answer on 41 of 46 frames for built banks, 15 for a dug-out channel, 38 for invasive plants and 30 for pipes |
| Flags the gate stopped                                                                                                                                    | not asked: an answer on the test is scored, never flagged                                                                                                                                                                        | 29 of 64 candidate flags dropped, 35 kept, because a model may flag only a feature it passed                                                                                    |
| One checker question about one frame, in cents                                                                                                            | not asked                                                                                                                                                                                                                        | Claude Haiku 4.5 0.2, Claude Sonnet 5 0.4, Claude Opus 5.5 1.0, Claude Fable 5.1 2.3                                                                                            |
| Cost per 100 frames, each asked the four questions three times by all four models, with the run's adversarial frames counted in                           | not asked                                                                                                                                                                                                                        | 51.3 USD, with direct calls at the full price                                                                                                                                   |
| What a phone downloads in the background on a first visit, so a test started online keeps going offline and the creek check works offline after one visit | 2.3 MB in 59 files, under the budget of 3 MB: the pages of the test and the creek check, and a phone-size copy of each of the 38 photos the test shows: the warm-up pair, the lesson and practice photos, and the 16 test photos | not asked                                                                                                                                                                       |

Which features each model passed on the 16-photo test: all four photos of a feature right in at least two of three runs (`results/model_pass_table.json`).

| Model            | Built bank | Dug-out channel | Invasive plant | Pipe running |
|------------------|------------|-----------------|----------------|--------------|
| Claude Haiku 4.5 | passed     | passed          | did not pass   | did not pass |
| Claude Sonnet 5  | passed     | passed          | did not pass   | did not pass |
| Claude Opus 5.5  | passed     | passed          | did not pass   | passed       |
| Claude Fable 5.1 | passed     | did not pass    | did not pass   | passed       |

The exact model ids, as sent to the API in the run of 2026-09-24: `claude-haiku-4-5-20251001`, `claude-sonnet-5`, `claude-opus-5-5` and `claude-fable-5-1`. Their prices and the day each was checked are in `docs/notes/model_ids.md`.

The full loop, from a desk: 3 video walks from 3 countries, each ending in a FHIR record made on the phone. The HL7 validator checked 17 records against OneAquaHealth's guide, 3 of them walk records, with 0 errors.

The test runs as a pre-registered study that stays open: it is the volunteer's own calibration step and the judges' first door, and a paid research panel may add sessions before the lock. No session from a person has arrived yet, so there is no human row here; sessions that arrive are reported once, after the lock, whatever they show.

Does the checker's question help? Part 2, the assisted second look, is analysed once, after the data lock, as tagged in `prereg-v2`.

### 3.2 The benchmark and its size

The same 16 photos, 4 per feature (2 with the feature, 2 without), run 3 times more by `evals/benchmark.py`, a second set of calls from the ones that wrote the pass table. Each cell is right answers of all answers, with the Wilson 95 percent interval in percent. Can't tell counts as wrong. The file is `results/benchmark_20260924T054939Z.json`.

| Model            | All 16 photos       | Built bank           | Dug-out channel      | Invasive plant    | Pipe running         | Can't tell |
|------------------|---------------------|----------------------|----------------------|-------------------|----------------------|------------|
| Claude Haiku 4.5 | 31 of 48 (50 to 77) | 12 of 12 (76 to 100) | 10 of 12 (55 to 95)  | 0 of 12 (0 to 24) | 9 of 12 (47 to 91)   | 35 percent |
| Claude Sonnet 5  | 33 of 48 (55 to 80) | 12 of 12 (76 to 100) | 12 of 12 (76 to 100) | 0 of 12 (0 to 24) | 9 of 12 (47 to 91)   | 27 percent |
| Claude Opus 5.5  | 33 of 48 (55 to 80) | 12 of 12 (76 to 100) | 9 of 12 (47 to 91)   | 0 of 12 (0 to 24) | 12 of 12 (76 to 100) | 31 percent |
| Claude Fable 5.1 | 34 of 48 (57 to 82) | 12 of 12 (76 to 100) | 10 of 12 (55 to 95)  | 0 of 12 (0 to 24) | 12 of 12 (76 to 100) | 27 percent |

Twelve answers per feature is small. An interval such as 55 to 95 percent means the test cannot tell a good model from a very good one. Read the intervals, not the point numbers, and read them as too narrow. Each interval counts the twelve answers of a feature, or all the answers of the whole test, as independent. They are not: the three runs repeat the same four photos of a feature, and a model mostly answers a photo the same way each time (the per-answer sweep of the same night, `results/model_sweep_20260924T054756Z.json`, shows it). Counted over the photos, each interval would be wider, so "each model beat the floor of always answering No" holds on the point numbers only.

This run kept counts, not replies. `make reproduce` checks that each accuracy and interval follows from the recorded counts, but the counts themselves, the Can't tell share and the count of malformed replies are as recorded: no reply is left to grade them from, and it prints a note saying so.

**On creek footage.** The four models also answered every feature question on 46 frames from 5 openly licensed videos in 3 countries. No frame has a label (0 labelled), so this measures agreement between models, not accuracy. The pair that agreed least, Haiku 4.5 and Fable 5.1, gave the same answer on 15 of 46 frames for a dug-out channel. Four frames drawn by code, a white one, a black one, a room made of flat rectangles and a screenshot of text (`evals/fixtures/__init__.py`), got can't tell as each model's majority answer over three runs, on every feature (`results/footage_latest.json`, under `adversarial`; the run kept the majority answers, not each reply).

### 3.3 How the models fail

- **A run once measured our config instead of the models.** The first paid sweep sent an answer tool whose allowed values came from YAML, where an unquoted yes and no are booleans. The models were told to answer `true` or `false`, and 73 of 144 answers were malformed. The values are now quoted and pinned by `evals/tests/test_model_sweep.py::test_the_answer_tool_allows_exactly_the_three_words_as_strings`; the run done again had 0 malformed. Both runs stay in `results/`.
- **No model passes invasive plants.** In the sweep behind the pass table, 47 of the 48 answers on the plant photos were can't tell. The plant photos are close views of a plant with no water in them, and the instruction says to answer can't tell when the photo does not show a stream. The notes show it: several name the plant, for example "possibly invasive Himalayan blackberry, but no stream is visible", and still answer can't tell. So this result measures our photos and our instruction as much as the models. The photos and the question wording are frozen with the tagged analysis plan, so changing them would be a deviation. The instruction is not in the plan: it is ours, in `evals/models.yaml`, and changing it would need a new paid run and a new pass table.
- **Can't tell is common.** Answers per feature in that sweep, all four models together:

| Feature         | Answers | Yes | No | Can't tell | Right |
|-----------------|---------|-----|----|------------|-------|
| Built bank      | 48      | 24  | 24 | 0          | 48    |
| Dug-out channel | 48      | 20  | 24 | 4          | 42    |
| Invasive plant  | 48      | 1   | 0  | 47         | 1     |
| Pipe running    | 48      | 22  | 23 | 3          | 43    |

- **Models disagree on footage.** On the same frames, Haiku 4.5 answered yes for a dug-out channel on a share of 0.2 of them and Opus 5.5 on a share of 0.0. Without labels we cannot say which is right.
- **A pass can be luck.** Four photos, three runs. A pass on a feature says the model got these four photos right, not that it sees that feature in general.
- **The note is checked for form, not for truth.** The gate refuses markup, control characters and long notes, but a note that is wrong in plain words gets through. That is why a flag can only ask the person to look again, and the person's answer is what is stored.

### 3.4 Cost

Every paid call is one line in `results/cost_log.jsonl`, with its tokens and its price; the fake runs spend nothing. All paid calls together: 5113 calls for 41.6 USD, of which footage 34.2 USD, the sweeps 3.7 USD and the benchmarks 3.2 USD. That includes the run that measured our config and the three-model run before Opus 5.5 and Fable 5.1 joined.

- The four-model run: the sweep 2.2 USD, the benchmark 2.2 USD and the footage 23.6 USD.
- Per 100 footage frames, four models, direct calls at the full price, with the run's adversarial frames counted in: 51.3 USD. The batch interface costs half as much, but a batch once waited three hours in the queue.
- The live site calls no model, so a volunteer's check costs nothing in model calls.

## 4. The FHIR mapping

The records one visit makes, and how the score reaches every Observation, are drawn in `docs/diagrams/fhir-graph.svg`. It is too wide to print here with text you can read; the list of what one visit produces, later in this section, names the same records.

OneAquaHealth says citizen data should stand beside lab data under the same profiles and value sets. A lab result is trusted because its quality checks travel with it. Second Look gives a volunteer's observation the same thing: their per-feature score, stored in the record of their test sitting beside a dated Practitioner qualification for the test, and linked through Provenance to every Observation they make. The four features are the ones the project lead named. The creek check asks the official Citizen Science App's questions in their order and in the app's own words and languages, quoted from its public translation bundle (`docs/notes/app_strings.md`). The city actions are OneAquaHealth's own restoration measures, from the [OneAquaHealth Policy Brief \(2026\)](#), page 9: replant margins, fix sewers, reconnect the floodplain, remove barriers, take out the concrete. The official app has no question for a dug-out channel, so the check asks none: that score is kept but stands beside no answer, and "reconnect the floodplain" waits for one; plants have no city measure of their own.

*Every surface of theirs we use, and where*

| Their surface                                       | What we use it for                                                                                                                                                                                                                                                                                                                               | Where                                                                                  |
|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| The official Citizen Science App's question wording | The creek check asks its questions word for word, in its order, with its closing question on feelings, each credited to the app and checked against its public bundle by <code>make app-strings-check</code>                                                                                                                                     | <code>content/form.yaml</code> , <code>content/app_strings.json</code>                 |
| The official app's own translations                 | The creek check speaks the official app's own languages (English, Portuguese, Dutch, Norwegian, French, Italian), so a volunteer in any pilot city sees the questions they already know. Each translation was read beside the English first; the 14 whose meaning differs show in English, marked, and the list is written up for the organizers | <code>/check</code> , <code>/walk</code> , <code>docs/notes/app_translations.md</code> |
| Their Location and Observation profiles             | Every spot and every answer                                                                                                                                                                                                                                                                                                                      | <code>core/fhir_emit.py</code> , <code>docs/fhir_mapping.md</code>                     |
| Their value sets and UCUM units                     | Present, absent, the indicator groups, metres                                                                                                                                                                                                                                                                                                    | <code>core/fhir_emit.py</code>                                                         |
| A Questionnaire through their form extension        | The test and the check, answered as <code>QuestionnaireResponses</code>                                                                                                                                                                                                                                                                          | <code>fhir/fsh/</code>                                                                 |
| Nested Locations                                    | Creek, reach and spot with part of                                                                                                                                                                                                                                                                                                               | <code>core/regions.py</code> , <code>content/regions/</code>                           |
| The HL7 validator with their guide, terminology on  | Sample records from both emitters in CI; golden vectors hold the live emitter to them                                                                                                                                                                                                                                                            | <code>scripts/fhir_validate.py</code> , <code>fhir/ig.lock</code>                      |
| Their sandbox                                       | Conditional creates with our tag and a ledger, and a Library entry for the data set                                                                                                                                                                                                                                                              | <code>scripts/repush_sandbox.py</code> , <code>fhir/sandbox_ledger.jsonl</code>        |
| Their decision tool's measures                      | What a creek needs, in their words, from the Policy Brief, page 9                                                                                                                                                                                                                                                                                | <code>content/approved_sentences.yaml</code> , <code>/city</code>                      |
| The five One Digital Health dimensions and FAIR     | Stated in words under One Digital Health and FAIR                                                                                                                                                                                                                                                                                                | this section                                                                           |
| The follower city recipe                            | <code>make new-city NAME=Aarhus COUNTRY=Denmark LAT=56.1629 LON=10.2039</code> scaffolds a new city in seconds; Heraklion was made that way, as a dry example                                                                                                                                                                                    | <code>scripts/new_city.py</code> , <code>docs/cities/</code>                           |
| Their SpecimenOah profile                           | The shape of a laboratory result coming back to a volunteer's pipe, marked EXAMPLE                                                                                                                                                                                                                                                               | <code>core/fhir_referral.py</code>                                                     |

| Their surface                                                                                                         | What we use it for                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Where                                             |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Not theirs: iNaturalist's public API, under its <a href="#">terms</a> ; each observation keeps its observer's licence | One context line per creek, research grade sightings of the region's listed invasive plants near its spots, shown once a finished check on that creek has answered the plant question, never in the check and never counted. The Bay Area list was approved for the team on 2026-09-25, and the daily job has stored sightings of listed plants near Strawberry Creek, but no finished check there has answered the plant question yet, so the line shows nothing so far. | <code>docs/adr/0011-inaturalist-context.md</code> |

One creek visit becomes these resources (`docs/fhir_mapping.md`):

| Resource                            | Profile                        | What it holds                                                                                                                                                                                                                             |
|-------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location, three nested              | their Location0ah              | creek, then reach (partOf creek), then spot (partOf reach, with position). Each has an identifier, a name and mode = instance, as the profile requires.                                                                                   |
| Practitioner                        | base R4                        | the volunteer, identified only by a random contributor token. One qualification coded second-look-test, period from the test date to 90 days later, issuer our Organization.                                                              |
| QuestionnaireResponse, test sitting | base R4                        | one item per feature with the score out of 4, computed by code. Authored by the Practitioner.                                                                                                                                             |
| QuestionnaireResponse, visit        | base R4                        | one item per answered check item. Link ids are the stable ids from <code>content/form.yaml</code> .                                                                                                                                       |
| Observation, one per answered item  | their ObservationIndicators0ah | status final, subject the spot, performer the Practitioner, effectiveDateTime, derivedFrom the visit response, a coded or quantity value.                                                                                                 |
| Provenance                          | base R4                        | target every Observation of the visit, agent author = Practitioner and assembler = our software Device, entity source = the visit response and the test sitting. This is the path from any answer to the score of the person who gave it. |
| Organization, Device                | base R4                        | us, and the software.                                                                                                                                                                                                                     |
| Bundle, type collection             | base R4                        | everything above. The emitter also produces a transaction Bundle with conditional creates for the sandbox mirror ( <code>to_transaction</code> in <code>core/fhir_emit.py</code> ).                                                       |

- `Observation.value`: their Temporary0ahSystem#present and #absent. "Can't tell" is our second-look#cant-tell because their system has no general unknown code.
- `Observation.category`: their indicator groups. Built banks and dug-out channel sit under #morphology (their spelling), invasive plants under #invasiveOrganisms, pipes and water height under #hydrology. The profile does not bind category, so this is additive.
- `Observation.code`: our feature codes artificial-bank, dug-out-channel, invasive-plant, pipe-running. Their profile binds code to 0ahIndicatorsNoHealth0ahVs with strength preferred, so a code outside it produces an information note, not an error. The validator confirmed this. Water height uses their own #hydrology code directly.
- Coding displays must be the code system's own display. The words a person sees on screen live in `content/form.yaml`, never in the coding.
- Units are UCUM (m for water height).

## One Digital Health and FAIR

*The five dimensions and the FAIR principles, one line each*

Story: a student walks to Strawberry Creek, takes a two-minute test on her phone, and from then on every observation she makes carries how well she sees each kind of damage. A city analyst reads her record beside a lab result under the same profile and knows how much weight to give each. The health card tells her one thing for herself, one for her dog and one her city could do.

Five dimensions, in words, because no script produces them: citizen engagement, strong; education, strong; human and veterinary healthcare, partial (approved sentences for the person and the pet, no diagnosis, no site risk); industry 4.0, partial (FHIR records, a gated vision model, an audit log); environment, strong.

FAIR: findable through a Library entry in their sandbox and a public repository; accessible through a read-only FHIR endpoint; interoperable through their profiles, value sets and UCUM; reusable through MIT code, a pinned guide, Provenance on every record and a tagged analysis plan.

## 5. Limitations

- Part 2 cannot show that the checker never misleads. Every one of the 6 flags in its set points the way of the gold label (`results/assist_flags.json`), so the question only ever follows a first answer that was wrong or Can't tell. Part 2 measures whether the question helps that person, and nothing more; a checker that is sometimes wrong would need a set with wrong flags in it.
- One labeller. The gold labels came from the picks file Alex wrote with the planner, a Claude chat (commit 81e62ed), and no second, blind label exists yet, so read every accuracy as agreement with this key (`docs/DATA_CARD.md`, `docs/deviations.md`).
- Four photos per feature is coarse: it shows a person what to practise and flags an answer worth a second look, but it is too coarse to weight votes by feature. In our simulation with made-up people (`results/consensus_coarseness.json`), weights from each feature's own photos did worse than a plain majority in all 5 skill patterns.
- A test started online keeps going if the network drops, from smaller copies of its photos. A reload while offline does not bring the sitting back: it opens the test at its start, or the offline page when the phone has not kept that exact link, such as `/t?src=other` opened once. The end of the lesson is not sent again once the network is back. The creek check works offline after one visit.
- The test and lesson photos are sent as the full-size JPEGs, so on a slow phone line each photo can take a few seconds to appear. That is on purpose: they are the study's pre-registered materials, and every person must see the same pixels, so everyone online gets them full size; only the landing page and the poster serve smaller copies to everyone. A phone also keeps a smaller copy of each test and lesson photo, shown only when a photo cannot be fetched at all, mid-sitting.
- The official app has no question for a dug-out channel, so the creek check asks none: that score is kept but stands beside no answer, and the city measure it would lead to waits for one. Plants that do not belong have no city measure of their own.
- Of the 35 footage flags the gate kept, 32 are on a dug-out channel, so they ask nothing. The other 3 are one model's runs on built banks, all on one frame of creek water over stones, seen from above, where the other models that passed built banks said can't tell. Passing four photos did not stop that flag, which is why a flag can only ask (`examples/footage-flag/`).
- The photos come from open collections in several countries and seasons, not from the creeks a Berkeley visitor will stand in.
- The footage labels come from the videos' own descriptions, and almost no openly licensed description names a feature, so the footage result leans on agreement between models, which is not accuracy.
- The citizen observer is modelled as a FHIR Practitioner, for want of a better fit in the guide; our proposal says so.
- The answer key can be rebuilt after the lock. From Sep 28, judge mode says right or wrong on the same 16 photos the test uses, and a retake in a fresh browser shows the score per feature each time, at any date. So a person who wants to can learn all 16 answers, take the test, keep 4 of 4 on every feature and carry that score into their creek checks, and nothing marks such a score. Before the lock judge mode is shut, so it could not leak the key while the study ran, and the study counts only the first finished sitting from a browser. A separate bank of photos for judge mode and for retakes would close it (`docs/THREAT_MODEL.md`).
- Their sandbox's name, `sandbox.h17europe.eu`, stopped resolving on Sep 23. `/two` can show their record only from a copy the Mac fetches while it answers, so for now it shows ours alone, and the re-push to their sandbox waits until it answers again.
- The dated and daily jobs run on one laptop, Alex's Mac, under his own logins: the copy of their record for `/two`, the iNaturalist line, the daily backup, the uptime check every 10 minutes, the daily OpenTimestamps stamp of the audit log, the sandbox re-push, the watch on our pull request and issues in their guide, and the one run of the analysis after the lock. While that laptop sleeps or is off, they wait. The live site keeps working without it, and the lock itself is the Worker's clock, not a job (`DEPLOY.md`, Jobs on the Mac).
- The tests step shows some tests as xfailed. Each is a known bug in our own code, in the gate, the FHIR writer, the referral check, the follow-up picker or the content loader, kept as a test of the right behaviour that fails today and is marked as a strict expected failure, so the run goes red the day it is fixed. Each is named, with what it means, in `docs/KNOWN_BUGS.md`.
- The tagged plan did not plan recruitment, and nobody has taken the test yet. The one finished sitting the public counts show, since Sep 25, is our own automated judge walk, which filled the hidden form field, so the plan's rules leave it out (`docs/deviations.md`). A paid research panel may add sessions before the lock if Alex launches it (a logged deviation); without it, whoever opens the link is whoever opens the link. The panel study has had no ethics review: it is an anonymous usability test that keeps no name or contact, and whether a paid one needs a review is checked before it launches.

- The AI numbers come from paid calls on Sep 23, Pacific time: four models, three runs each, on 16 photos, once for the pass table and once more for the right-answer counts, plus the footage run. That is small. Read the intervals in the benchmark file, not the point numbers, and read them as too narrow: they count each run's answer as new, but the three runs repeat the same 16 photos and a model mostly answers a photo the same way each time ([results/model\\_sweep\\_20260924T054756Z.json](#) has every answer), so the answers are not independent. Counted over photos, each interval would be wider.
- English only. A Spanish draft exists and stays out of the build until a fluent person signs it.

## 6. Contributions

1. **A measured observer, carried in the record.** A two-minute test gives each volunteer a score per feature, and OneAquaHealth's own profiles carry it, through a qualification and Provenance, to every Observation that person makes. A city analyst reads each answer beside how well its observer sees.
2. **A model that may ask and never decide.** A model is licensed feature by feature by the same test people take, from a committed pass table. Its output becomes a flag through one gate or is dropped, and the record is built from human inputs only.
3. **Measurement that shows its failures.** The pass table, the benchmark with its intervals, the run that measured our own config, and the plant photos that no model could judge are all in `results/`, with the cost of every call.
4. **Open work, and work given back.** The code is MIT. What we contributed back, as the README lists it:

Sent to OneAquaHealth's implementation guide on 2026-09-24, in the open:

- [hl7-eu/oah pull request 5](#): our citizen observer example, five FSH files that build inside their guide with SUSHI 3.20.1 at no errors, and a proposal for carrying observer quality in their guide ([docs/ig\\_proposal.md](#)), with the gaps our validator runs found. It asks which resource should stand for a citizen observer.
- [hl7-eu/oah issue 6](#): their temporary code system spells one code morphology. We kept their spelling so our records validate.
- [hl7-eu/oah issue 7](#): `SpecimenOah.collection.collector` allows only a `PractitionerRole`, which leaves out a laboratory and a volunteer who takes a sample.
- [hl7-eu/oah issue 8](#): their sandbox's name stopped resolving on 2026-09-23, with the evidence from their own nameserver.

And for anyone, not sent to them: a read only MCP server over our own records, so any software agent can ask for a creek's records with the resource ids behind every answer ([examples/mcp/README.md](#)).

## 7. How to check this report

- `make judge-check` runs the tests with no key and no network.
- `make verify-claims` checks every number in the README and the docs against `results/`.
- `make report-pdf` builds this file again from its sources; a test fails when the PDF is older than the README sections and results it was built from.
- The model card, the data card and the threat model are `docs/MODEL_CARD.md`, `docs/DATA_CARD.md` and `docs/THREAT_MODEL.md`.

AI coding tools wrote most of the code and the text, from written briefs, and every change was checked by `make check` before it was committed. The people set the direction and made every decision that needs a person, as the README says under "How this was built".